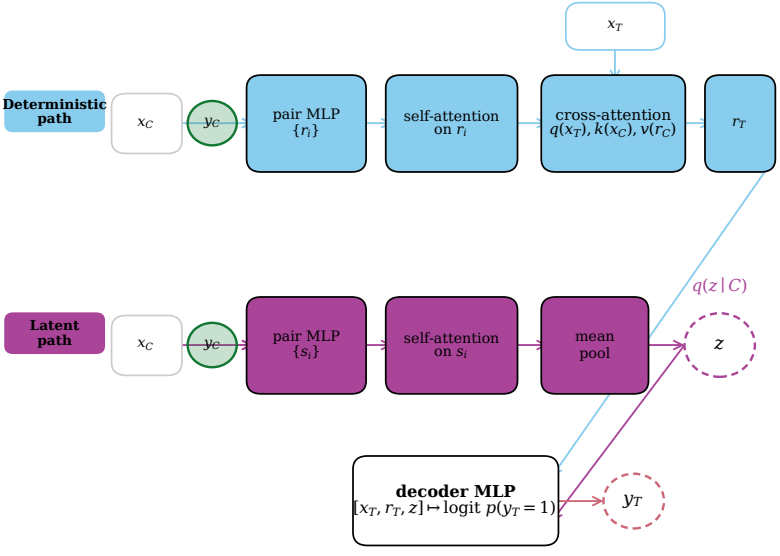
Attentive Neural Process (paper architecture)

Self-attention appears separately on the deterministic and latent paths. Targets attend to labelled contexts only on the deterministic path.

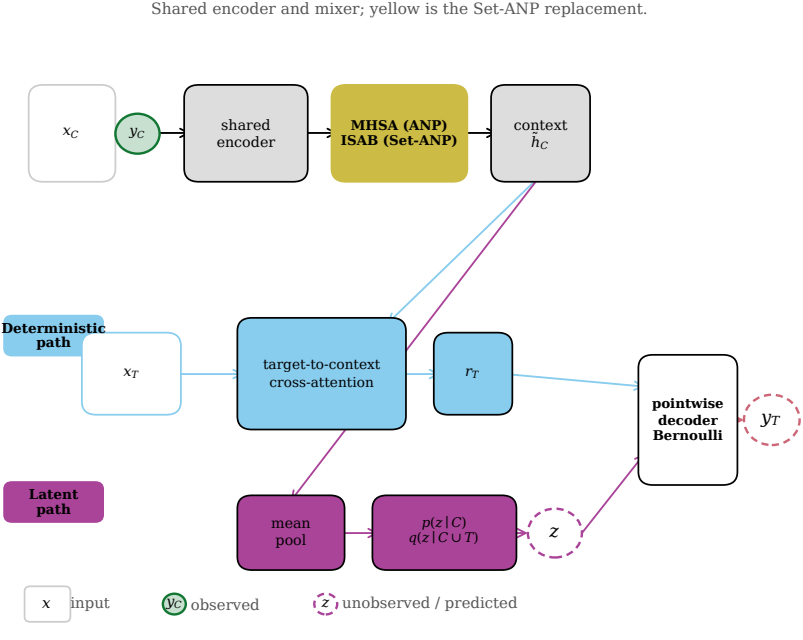


At training, the latent encoder also forms $q(z \mid C \cup T)$.

x input y_C observed z unobserved / predicted

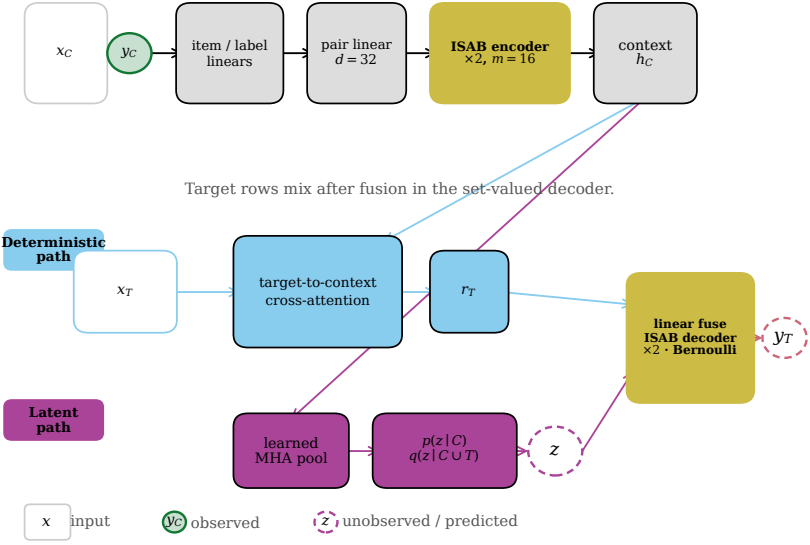
Our ANP and the clean Set-ANP ablation

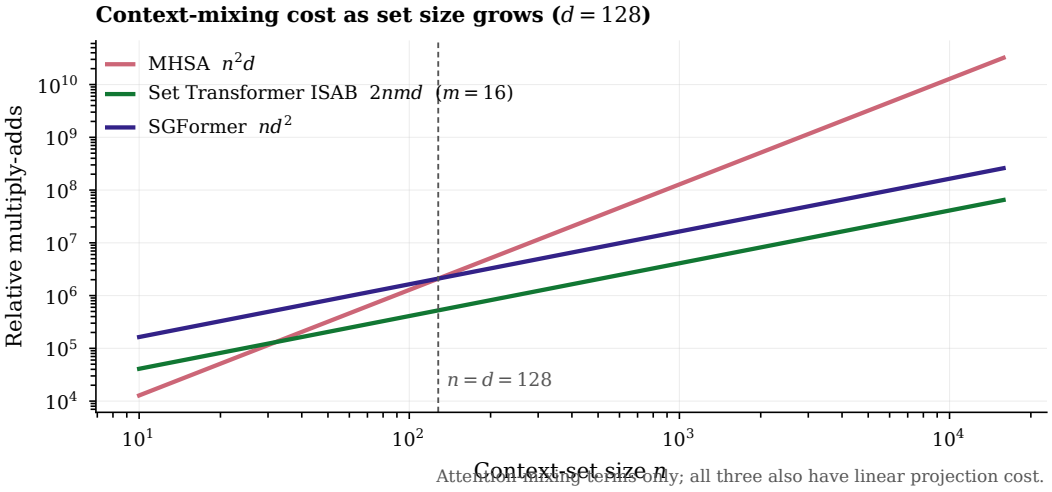
One shared labelled-context encoder feeds both paths. Set-ANP changes only the one context-mixing block.



AEA-NP: full Set-Transformer latent Neural Process

This is not the one-block Set-ANP ablation. It also changes latent pooling and lets target rows mix in the decoder.




Context-mixing cost as set size grows ($d = 128$)
MHSA n^2d
Set Transformer ISAB $2nmd$ ($m = 16$)
SGFormer nd^2
Relative multiply-adds
10^{10}
10^{9}
10^{8}
10^{7}
10^{6}
10^{5}
10^{4}
$n = d = 128$
10^{1}
10^{2}
10^{3}
10^{4}
Context-set size n
Attention mixing terms only; all three also have linear projection cost.